

PH 702: Advanced Quantum Mechanics

Course Code	PH 702
Course Name	Advanced Quantum Mechanics
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	PhD
Distribution	Elective
Approval	2nd Adhoc Senate (May 2012)

Course Contents

- **Module I**
 - Review of basic concepts in quantum mechanics, measurements, observables and generalized uncertainty relations, change of basis, generator of translation [3 Lectures]
- **Module II**
 - Rotations and angular momentum commutation relations, spin-1/2 systems and finite rotations of general theory of angular momentum, addition of angular momenta, Clebsch-Gordon Coefficient (CGC) and its properties, conventions of CGC, choice of phases and 3-j symbols, Schwinger's oscillator model of angular momentum [4 Lectures]
- **Module III**
 - Tensor operators, Cartesian tensors, irreducible tensor operators, matrix elements of tensor operators, Wigner-Eckart theorem and its applications [4 Lectures]
- **Module IV**
 - Symmetry principles in quantum mechanics, conservation laws and degeneracies, discrete symmetries, parity and time reversal, symmetry in collision and ionization process [4 Lectures]
- **Module V**
 - Approximation methods in quantum mechanics, review of non-degenerate perturbation theory, degenerate perturbation theory, time dependent perturbation theory, periodic perturbation, Rabi flopping frequency, Adiabatic approximation, Berry phase, sudden approximation [5 Lectures]
- **Module VI**
 - Interaction of quantum systems with radiation, Dipole approximation and selection rules, length, velocity and acceleration forms of matrix transition matrix elements, Absorption process, spontaneous emission, stimulated absorption, Einstein coefficients, life times, line intensities, widths and shapes [5 Lectures]
- **Module VII**
 - Many electron atoms, central field approximation, self-consistent field method, Hartree-Fock equations, Koopman's theorem [5 Lectures]
- **Module VIII**
 - Relativistic quantum mechanics, Klein-Gordon equation, Dirac equation, probability densities and current densities, plane wave solutions of Dirac's equation, solutions of the Dirac equation for a central potential, non-relativistic limit of Dirac equation, negative energy states and hole theory [5 Lectures]
- **Module IX**
 - Second quantization, creation and destruction operators, occupation numbers, commutation relations, matrix elements of H for N-electron systems, field operators [5 Lectures]

Text and Reference Books

1. J.J. Sakurai, **Modern quantum mechanics**
2. Tannoudji, B Dui, F Laloe, **Quantum Mechanics** -Vol. I & II, Cohen.
3. Messiah, **Quantum Mechanics**, Vol. I and II
4. Bransden and Joachain, **Quantum mechanics**
5. Bransden and Joachain, **Physics of atoms and molecules**
6. R Shankar, **Principles of quantum mechanics**.
7. S. Raimes, **Many electron theory**.
8. Bethe, **Intermediate quantum mechanics**.
9. Landau & Lifshitz, **Quantum mechanics**.